

Multivariate Socio-Economic Analysis of Deprivation in Ireland

Valeria Garcia Hernandez

Overview

This project applies a range of multivariate statistical and machine learning techniques to analyze socio-economic data and evaluate the presence of underlying structure within the data.

The analysis is based on the 2022 Pobal HP Deprivation Index dataset. Based on the 2022 Census of Population, the 2022 Pobal HP Deprivation Index shows the level of overall affluence and deprivation at the level of electoral division across Ireland.

The primary objective of this project is to explore relationships between socio-economic variables, identify potential patterns through clustering, and evaluate how well these patterns align with recorded deprivation measures. In addition, the project investigates the predictive capability of classification methods in distinguishing areas of higher and lower deprivation, and examines dimensionality reduction techniques to better understand the structure of the data. Through a combination of exploratory data analysis, clustering, classification, and regression techniques, this study aims to provide both statistical insight and interpretable conclusions about socio-economic variation in Ireland.

Describing the Data

The 2022 Pobal HP Deprivation Index dataset contains 3417 rows and 17 columns. The first two columns contain geographic identifiers. Columns 3-5 contain the socio-economic variables to be analyzed. Columns 16-17 contain the deprivation index score and category, i.e., `index_lab` = “Extremely Disadvantaged” if `index_rel` ≤ -30 , “Very Disadvantaged” if `index_rel` $\in (-30, -20]$, “Disadvantaged” if `index_rel` $\in (-20, -10]$, “Marginally Below Average” if `index_rel` $\in (-10, 0]$, “Marginally Above Average” if `index_rel` $\in (0, 10]$, and “Affluent” if `index_rel` > 10 , where `index_rel` is estimated and provided by Pobal.

Feature Summary

ED_ID	Electoral district ID
ED_NAME	Electoral district name
TOTPOP	Census total population 2022
AGEDEP	% population aged under 15 or over 64 years of age
EDLOW	% population with a primary school education only
EDHIGH	% population with a third level education
HLPROF	% households headed by professionals or managerial and technical employees
LCLASS	% households headed by semi-skilled or unskilled manual workers
PEROOM	Mean number of persons per room
LONEPA	% households with children aged under 15 years and headed by a single parent
UNEMPM	% male unemployment
UNEMPF	% female unemployment
OWNOCC	% accommodation owned
PRRENT	% accommodation privately rented
LARENT	% local authority rented accommodation
index_rel	relative deprivation index score
index_rel	deprivation index label

EDA and Data Preparation

We visualize each socio-economic variable using histograms to examine their marginal distributions.

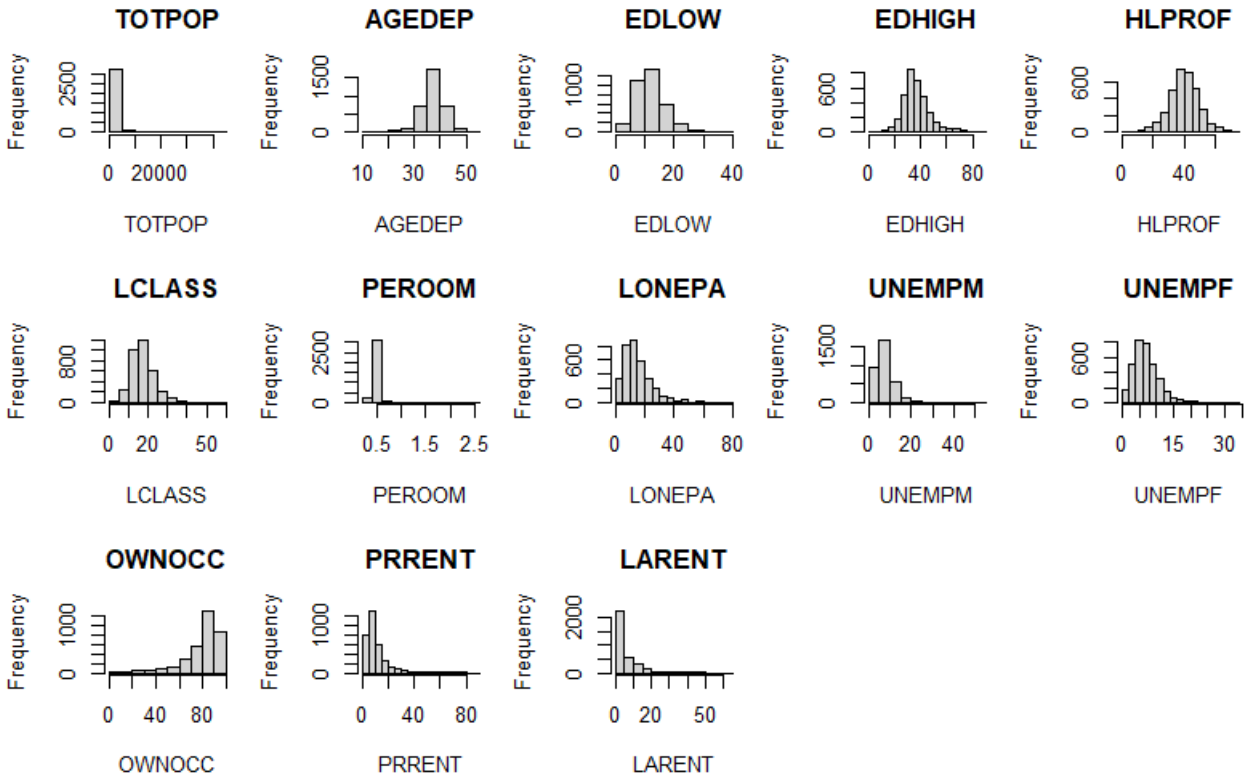


Figure 1: Socio-economic variables distribution

The histograms show that several socio-economic variables, such as TOTPOP, PEROOM, UNEMPM, UNEMPF, PRRENT, and LARENT, exhibit strong right skewness. This indicates the

presence of extreme values and potential outliers, which can disproportionately influence distance calculations in clustering methods such as k-means and hierarchical clustering, leading to clusters driven by outliers rather than meaningful structure. In particular, k-means is sensible to outliers due to its reliance on the mean, which is pulled toward extreme values. Similarly, in Principal Component Analysis (PCA), outliers can pull the principal components toward them.

From a socio-economic perspective, these extreme values may correspond to a small number of regions with unusually high population density, unemployment, or reliance on rental accommodations. This suggests that deprivation may be unevenly distributed across electoral divisions in Ireland.

The variables are also measured in different scales, with some variables such as TOTPOP taking very large values while others are expressed as percentages. As clustering methods rely on distance calculations, variables with larger magnitudes can dominate the analysis. **We standardize the data by subtracting the mean and dividing by the standard deviation of each variable,** ensuring that all variables contribute equally to the clustering solution.

We then use scatter plot matrices to explore relationships and potential correlations between variables.

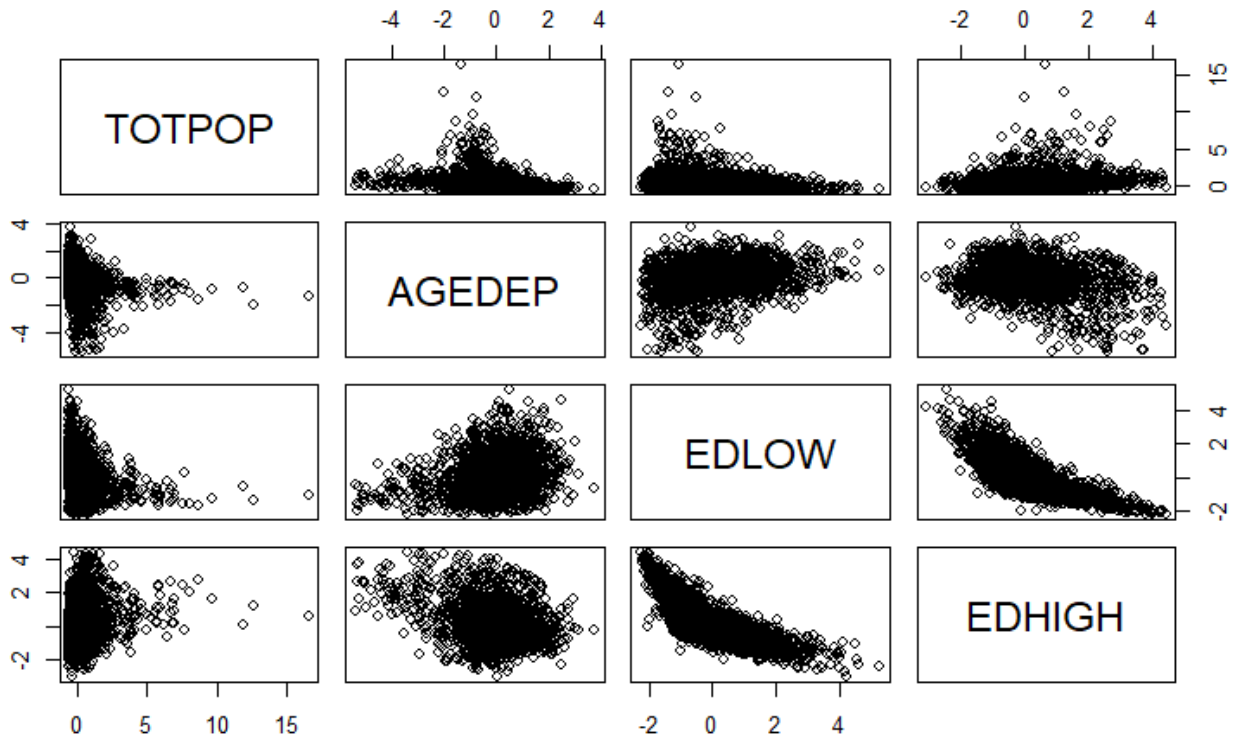


Figure 2: Scatter plot matrix 1

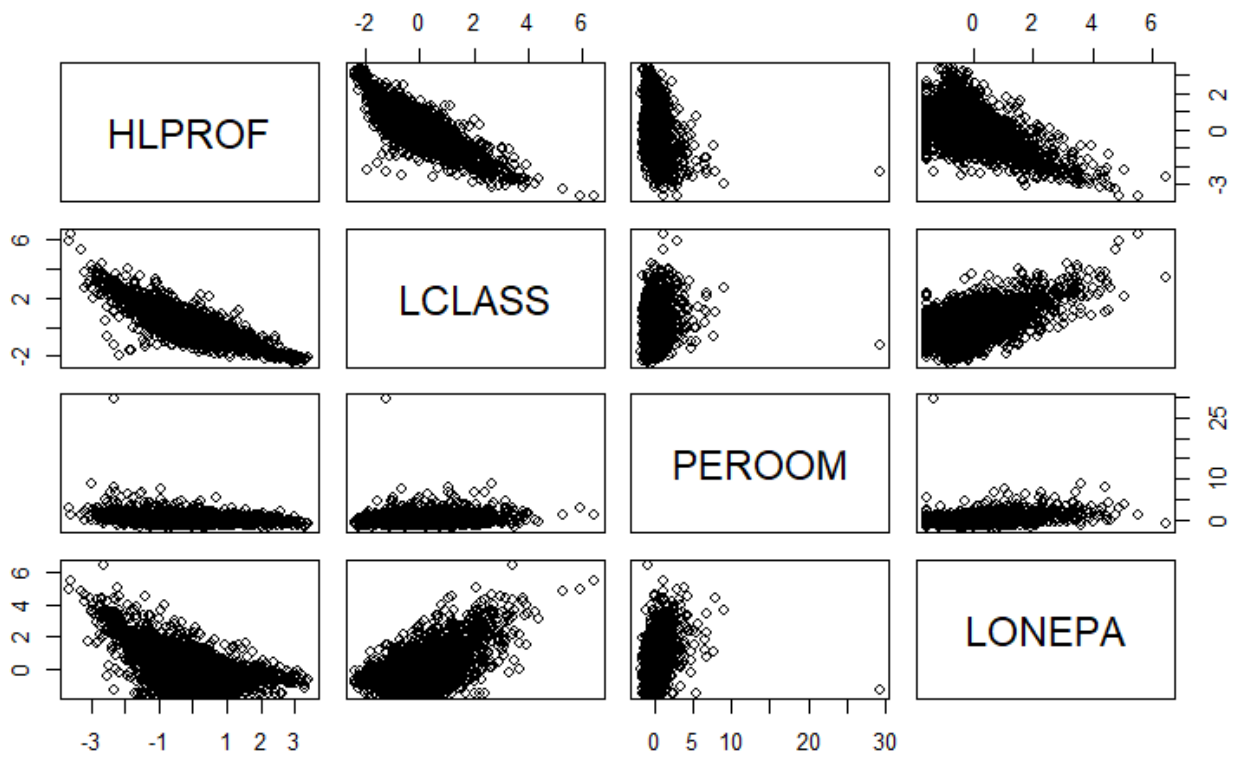


Figure 3: Scatter plot matrix 2

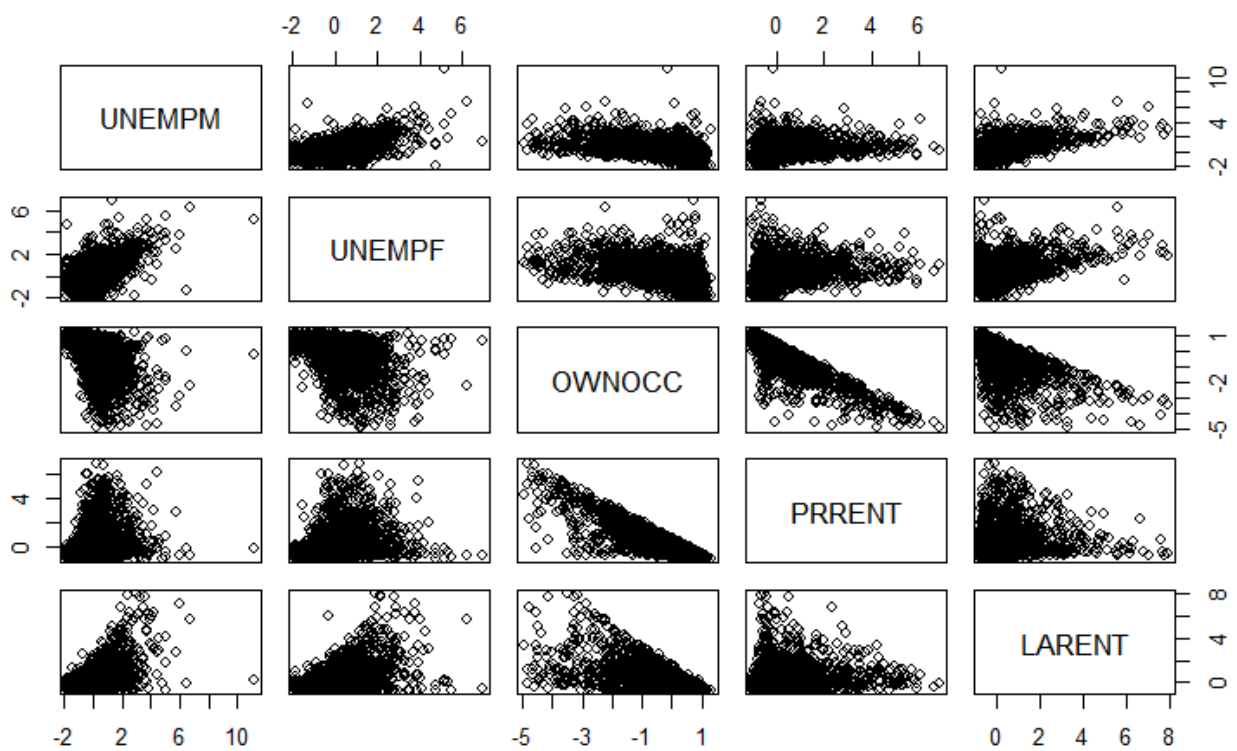


Figure 4: Scatter plot matrix 3

From the graphs, there is evidence of correlation between variables, such as EDLOW and ED-HIGH, HLPF and LCLASS, OWNOC and PRRENT, and between male and female unemployment, which may introduce redundancy. In such cases, dimensionality reduction techniques, such as PCA, could be useful.

Hierarchical Clustering

We perform hierarchical clustering. Since all variables have been standardized to a common scale, Euclidean distance is used as the dissimilarity measure. For the linkage method, we use average linkage, as it provides a balance between forming compact clusters and avoiding the sensitivity to extreme distances seen in complete linkage.

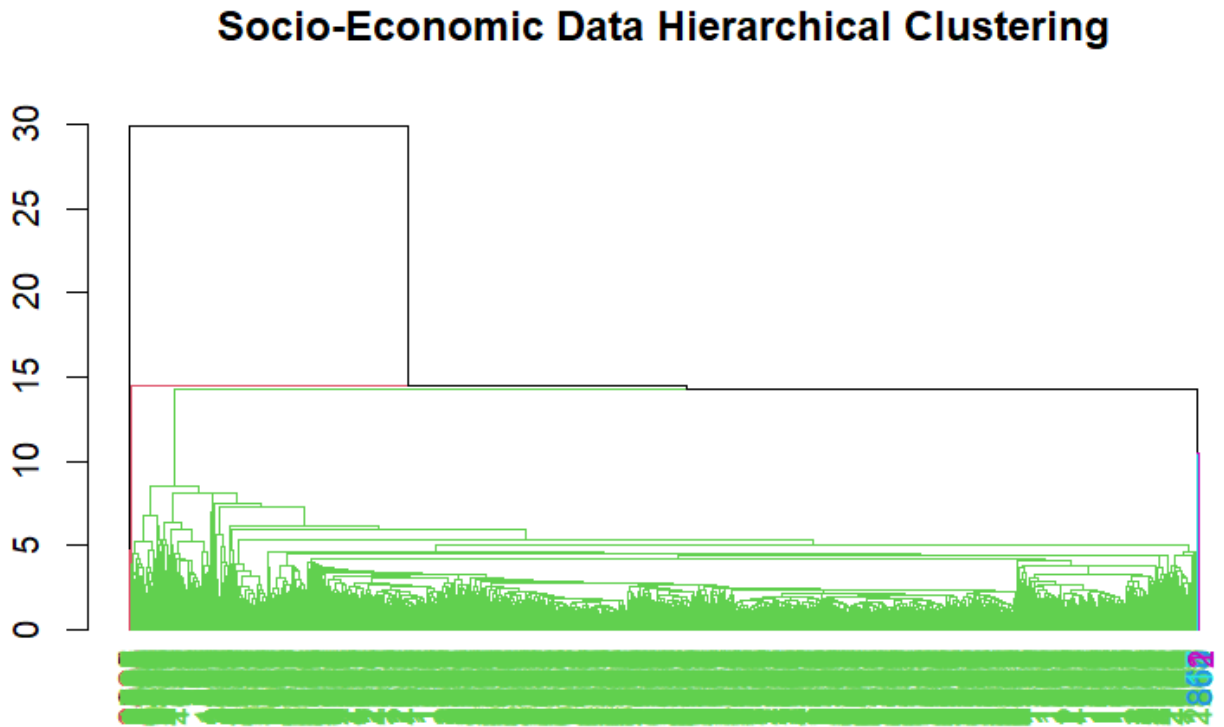


Figure 5: Hierarchical clustering dendrogram

From the dendrogram, there appear to be two main clusters. When compared to the six deprivation index categories provided by Pobal, this division may seem overly simplistic and not aligned with the labels. However, these clusters may instead capture a broader underlying structure in the data, potentially representing two general groups such as relatively advantaged and disadvantaged areas.

We cut the dendrogram into six clusters, however, as the goal is to recover the underlying socio-economic structure represented by the six deprivation index labels. We then calculate the Adjusted Rand Index (ARI) to quantify the degree of agreement between the clustering solution and the labels. Since $ARI = 0.0026$ is very close to 0, this indicates that the agreement is no better than random. Therefore, the hierarchical clustering solution does not recover the socio-economic structure captured by the deprivation index labels.

K-Means

We now perform k-means clustering. We run the k-means algorithm over the range of values $k=1$ to 10 and record the within group sum of squares for each value of k .

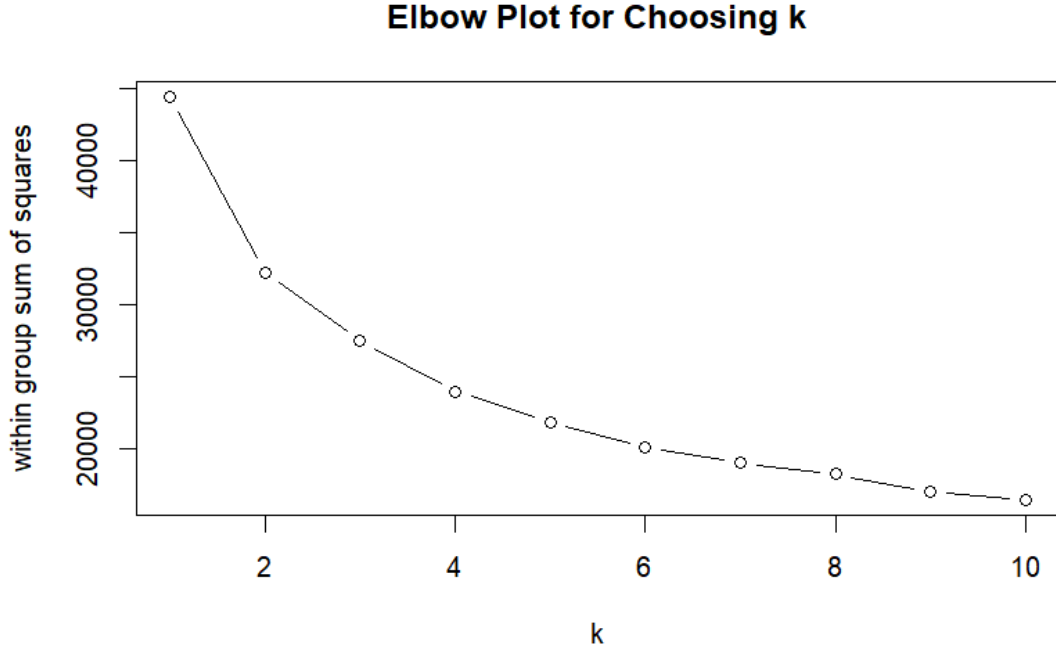


Figure 6: Plot of the within sum of squares for different values of k

From figure 6, $k = 4$ seems like a reasonable choice based on how much the within group sum of squares decreases as we increase k . However, we consider $k = 6$ since there are six different deprivation index labels, allowing for a direct comparison between this clustering solution and the known socio-economic groups.

For $k = 6$, we run the algorithm 10 times with different random starting points and keep the solution with the lowest within-cluster variation to avoid convergence to a local minimum. The ARI between the deprivation index labels and our k-means clustering solution is 0.31, indicating a moderate agreement between the two. While our k-means clustering solution is picking up some real structure, many data points are still grouped differently than the labels.

Since the k-means clustering solution has the higher ARI, it performs better in recovering the structure of the true deprivation index labels and is therefore preferred among both clustering methods.

Stability of K-Means: We want to assess whether the k-means clustering solution is stable and generalizable. First, we split the data into a calibration (70%) and validation (30%) sample and run k-means on the calibration sample. We then assign each validation observation to its nearest centroid from the calibration solution (call it S_1). Finally, we independently cluster the validation sample into six groups (call it S_2) and compare the two clustering solutions.

Note that we use the original data when splitting, compute the scaling parameters (mean and standard deviation) using only the calibration data, and then use those parameter to scale both sets. This is done to avoid data leakage.

The resulting $ARI = 0.39$ indicates moderate agreement between S_1 and S_2 . This suggests that while the clustering structure is partially stable, there is noticeable variation depending on the

sample used and the solution is only moderately generalizable. The clustering solution captures some underlying socio-economic structure, but the boundaries of the clusters are not clearly defined.

Binary Classification

We derive the bi variate categorical variable `index_lab_simp` where `index_lab_simp = 1` if above average deprivation index and `index_lab_simp = -1` if below average deprivation index.

LDA: We apply LDA using the standardized socio-economic data to predict `index_lab_simp`, assuming both classes share the same covariance matrix. To assess performance, we use leave-one-out cross-validation and compare the classes predicted by the model to the true values of `index_lab_simp`. The **accuracy** rate is 0.93, meaning that the model correctly predicts `index_lab_simp` for 93% of the observations. Since the two classes, -1 and 1, in `index_lab_simp` are roughly balanced, accuracy is an appropriate performance measure. This suggests that there exists a strong linear boundary between above average deprivation and below average deprivation areas based on the variables considered.

Note that the data is standardized prior to fitting the LDA model to account for differences in scale, introducing a small degree of data leakage.

Considering QDA: The Quadratic Discriminant Analysis (QDA) model's accuracy is 0.886. Since the QDA accuracy is lower, LDA is preferred for its simplicity.

Principal Component Analysis

We perform PCA on the standardized socio-economic data. We use the standardized data since PCA is sensitive to scale differences across variables.

We calculate the proportion of variation in the data accounted for by each principal component (PC) and use a scree plot to determine how many components are needed to represent the data.

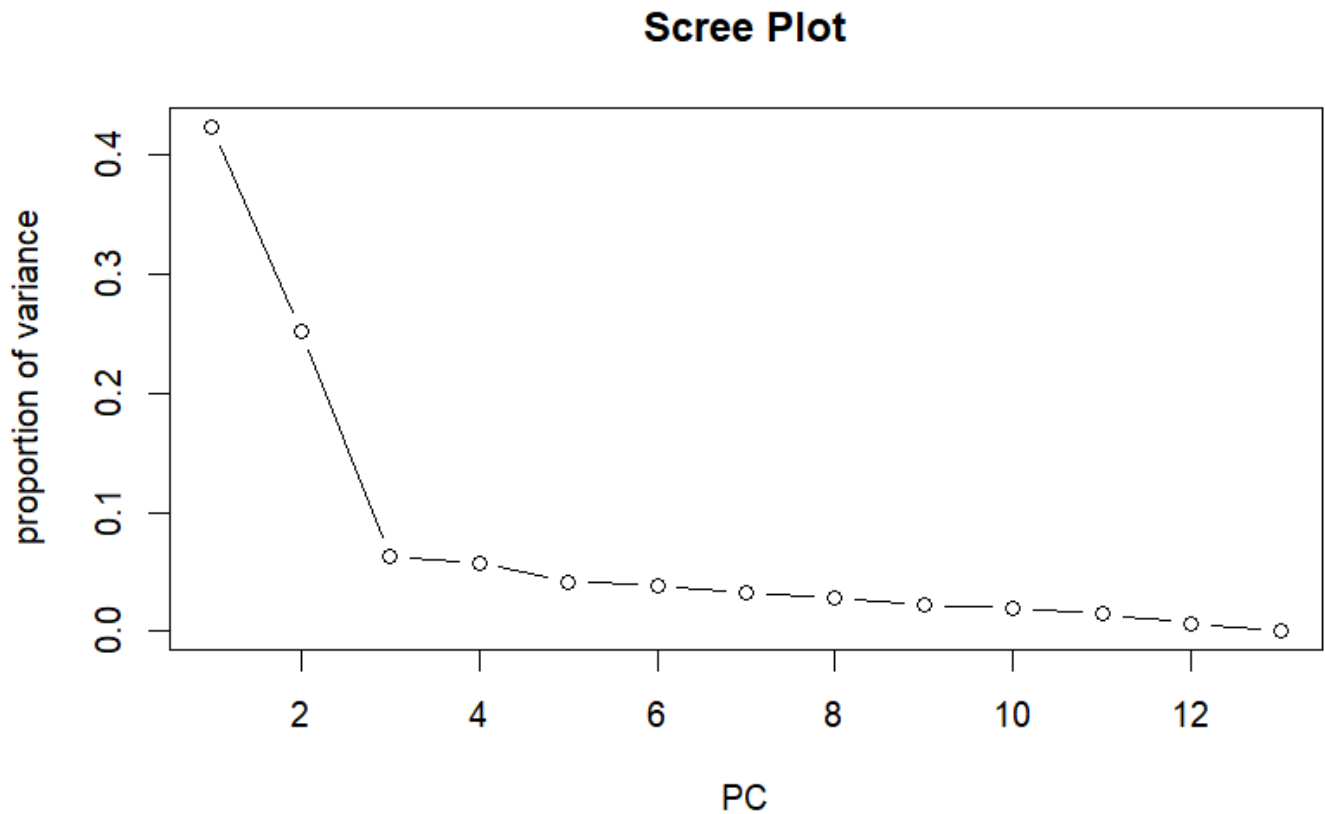


Figure 7: Scree plot

From the scree plot, we observe an “elbow” at the third principal component, after which the proportion of variance explained by the PC stays really low. The proportion of variance explained by PC1, PC2, and PC3 is 0.4231, 0.2517, and 0.06295, respectively. Therefore, we retain the first three principal components as they capture the majority of the variation in the data.

We first plot the first three PC scores:

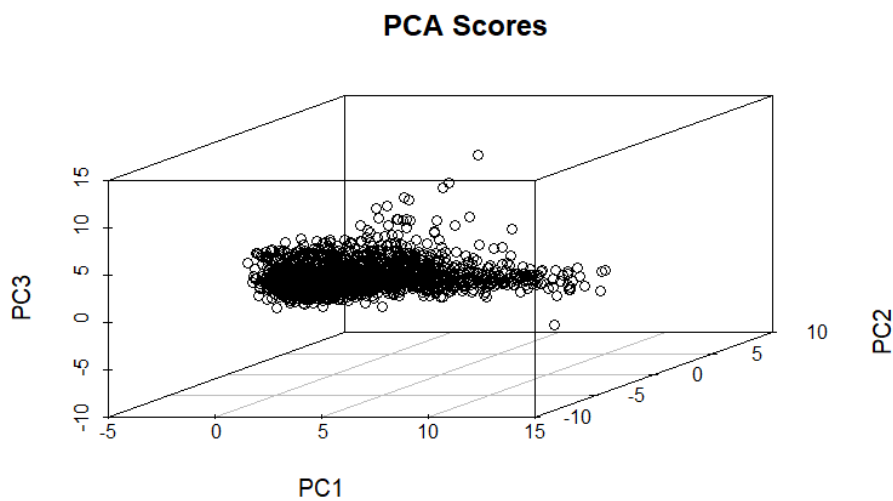


Figure 8: Three-dimensional plot of the first three principal components

We can also plot only the first two PC scores, as they capture approximately 70% of the total

variation in the data.

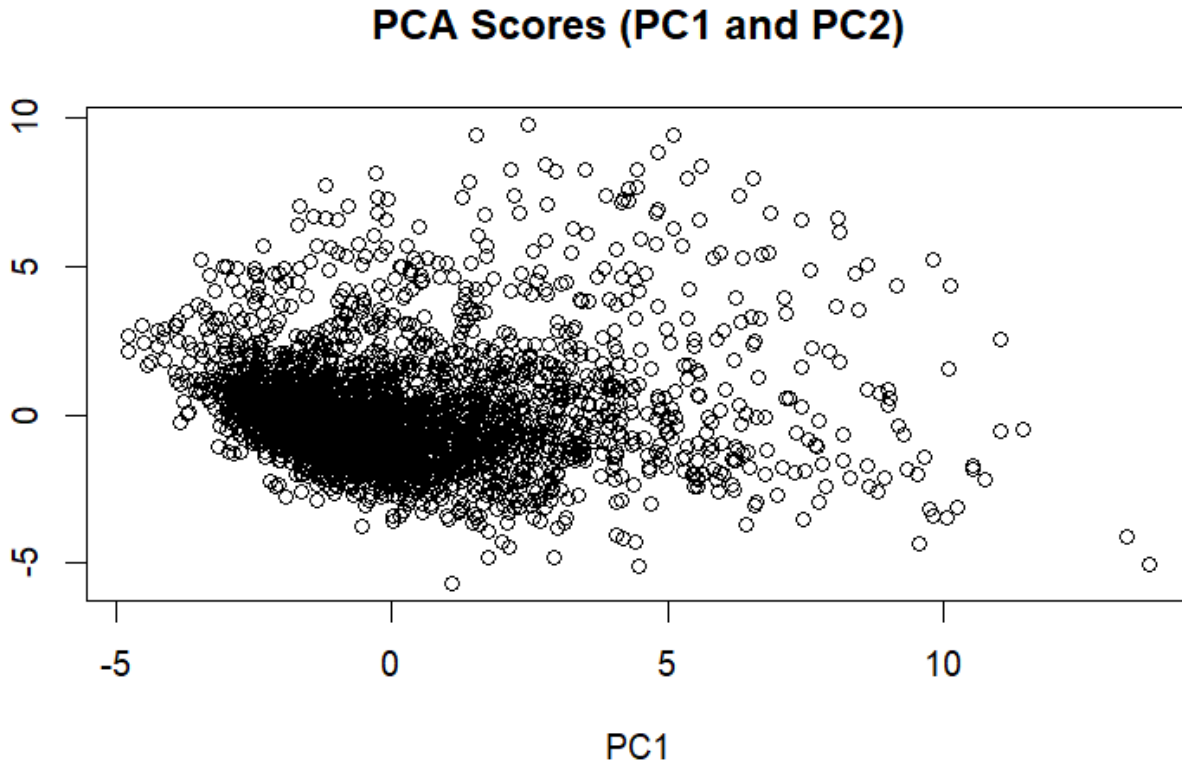


Figure 9: Two-dimensional plot of the first two principal components

The PCA scores plots show that the observations are concentrated in a relatively dense region, with no clear separation into distinct clusters. This suggests that clusters are likely to overlap. This is consistent with what we observed in the hierarchical clustering dendrogram, where there appeared to be two main clusters, one of which was significantly larger than the other.

K-Means on PCA: We cluster the PCA-reduced data using three PCA scores and k-means. The ARI between the deprivation index labels and the PCA k-means clustering solution is 0.346, which is slightly higher than the $\text{ARI} = 0.31$ for the original variables.

Clustering on the original variables uses all available features, preserving the full structure of the data. In contrast, clustering in a reduced-dimension space is performed on a smaller number of transformed variables that aim to capture the main variation in the data. This approach is based on principal components, which are uncorrelated and ordered by how much variation they explain. Clustering in a reduced-dimension space can reduce noise and redundancy, handle multicollinearity, improve computational efficiency, and make visualization easier due to the smaller number of variables. However, one disadvantage is that the clustering solution becomes more difficult to interpret, since clusters are defined in terms of principal components rather than the original variables.

In this case, PCA provides only a slight improvement in clustering performance, suggesting that it preserves the relevant structure of the data while reducing dimensionality. However, the difference is small and the solution becomes more difficult to interpret.

LDA on PCA: We refit LDA to predict `index_lab_simp` using the first two principal components and plot the resulting linear Bayes decision boundary.

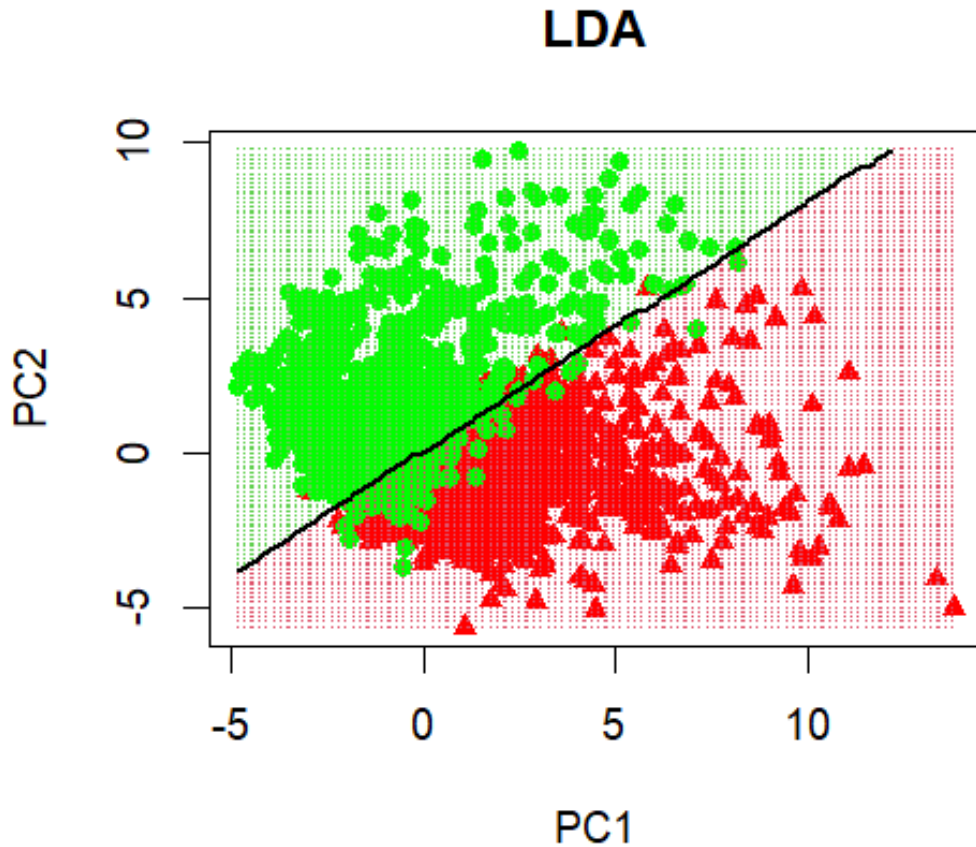


Figure 10: Plot of PCA scores and the linear Bayes decision boundary

From the plot, the linear decision boundary appears to separate the below average and above average deprivation areas effectively, suggesting that the two groups are linearly separable.

Principal Component Regression

We fit Principal Component Regression (PCR) to the data using cross-validation. To avoid data leakage, we pass the original (unscaled) data to the `pcr` function and set `scale = TRUE`, ensuring that centering and scaling are performed within each segment of cross-validation.

To evaluate the performance of the PCR model, we plot the cross-validated mean squared error for different numbers of components. The mean squared error of prediction (MSEP) decreases rapidly for the first two components and then stabilizes. Based on this, we select two components as a suitable trade-off between model complexity and predictive performance.

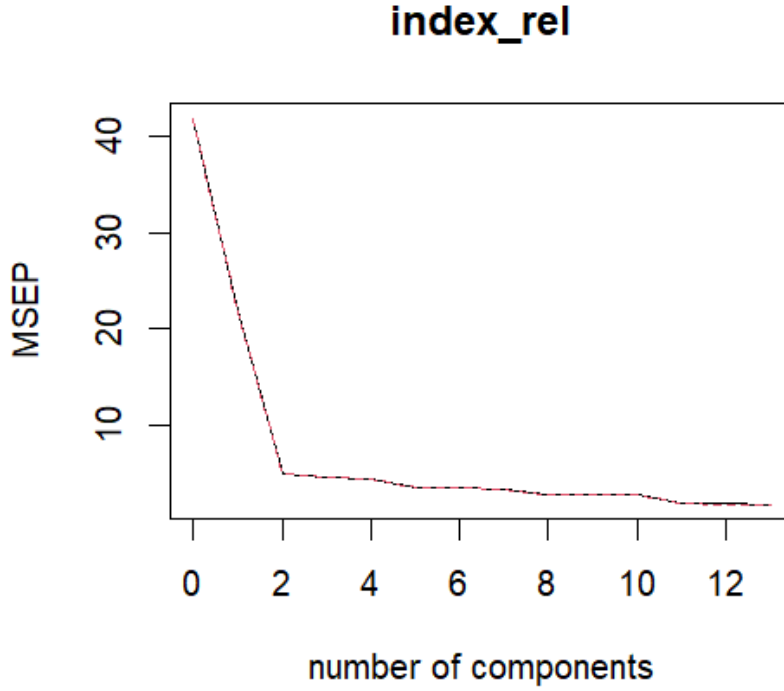


Figure 11: Plot of MSEP for the different numbers of components

The model's MSEP for the two components is 4.94, corresponding to an RMSEP of 2.22. This indicates that, on average, the model's prediction error is 2.22 units. Thus, the model can predict deprivation index values with a moderate average error.

PCR is useful because it reduces the number of predictors used in the model by projecting the data onto a set of uncorrelated principal components, which helps in addressing multicollinearity. This can improve model stability and predictive performance. However, PCR can also be more difficult to interpret since the predictors are now linear combinations of the original variables.

Insights

Overall, the analysis reveals that the socio-economic variables contain meaningful information about deprivation, but the structure is not captured equally well by all methods. The LDA model achieves high classification accuracy, indicating that a strong linear boundary exists between above average and below average deprivation areas. This suggests that the variables are effective at differentiating between deprivation categories when defined in a binary setting.

In contrast, the clustering methods show weaker alignment with the provided six deprivation index labels, with low ARI values for both the original and PCA-reduced data.

The results suggest that while socio-economic variables have strong predictive power for binary deprivation classification, they do not fully capture the underlying structure of the deprivation categories defined by Pobal.